Example 2.19

Show that $|3z-5+i|=4$ represents a circle, and, find its centre and radius.

Solution

The given equation $|3z-5+i|=4$ can be written as

$$3\left|z-\frac{5-i}{3}\right| = 4 \Rightarrow \left|z-\left(\frac{5}{3}-\frac{i}{3}\right)\right| = \frac{4}{3}.$$

It is of the form $|z-z_0|=r$ and so it represents a circle, whose centre and radius are $\left(\frac{5}{3},-\frac{1}{3}\right)$ and $\frac{4}{3}$ respectively.

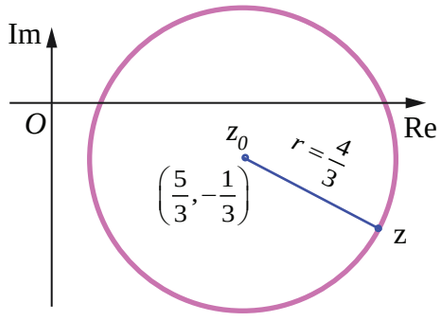


Fig. 2.24

Example 2.20

Show that $|z+2-i|<2$ represents interior points of a circle. Find its centre and radius.

Solution

Consider the equation $|z+2-i|=2$.

This can be written as $|z-(-2+i)|=2$.

The above equation represents the circle with centre $z_0=-2+i$ and radius $r=2$. Therefore $|z+2-i|<2$ represents all points inside the circle with centre at $-2+i$ and radius 2 as shown in figure.

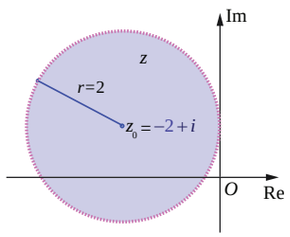


Fig. 2.25

Example 2.21

Obtain the Cartesian form of the locus of z in each of the following cases.

(i) $|z|=|z-i|$ (ii) $|2z-3-i|=3$

Solution

(i) we have $|z| = |z-i|$

$$\Rightarrow |x+iy| = |x+iy-i|$$

$$\Rightarrow \sqrt{x^2+y^2} = \sqrt{x^2+(y-1)^2}$$

$$\Rightarrow x^2+y^2 = x^2+y^2-2y+1$$

$$\Rightarrow 2y-1 = 0.$$

(ii) we have $|2z-3-i| = 3$

$$|2(x+iy)-3-i| = 3.$$

Squaring on both sides, we get

$$|(2x-3)+(2y-1)i|^2 = 9$$

$$\Rightarrow (2x-3)^2+(2y-1)^2 = 9$$

$\Rightarrow 4x^2+4y^2-12x-4y+1 = 0$, the locus of z in Cartesian form.

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